

MiniVault: A Crypto-Collateralized Stablecoin with a Same-Block Oracle Circuit Breaker

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ABSTRACT — This paper presents *MiniVault*, a minimal crypto-collateralized stablecoin protocol implemented in Solidity and deployed to the Ethereum Sepolia testnet. Users deposit ETH as collateral and mint *mUSD*, a USD-pegged ERC-20 token, against it at a 150% minimum collateralization ratio. The protocol is built from three contracts — an ERC-20 stablecoin, a Chainlink-fed oracle adapter, and a CDP vault — and is exercised through 33 passing unit tests covering happy paths, collateralization invariants, and liquidation. The principal contribution is a *same-block circuit breaker* embedded in the oracle adapter that rejects price updates moving more than 5% within a single block. This defense is motivated by the October 2022 Mango Markets exploit, in which \$114M was drained through flash-loan-driven oracle manipulation. I discuss the breaker's mechanism, the EVM rollback semantics that constrain its persistence, and the design trade-offs between false-positive volatility and security. The full system is live on Sepolia with verified source code on Etherscan.

1. Introduction

Stablecoins occupy the load-bearing role in decentralized finance: they are the unit of account in which lending, trading, and settlement are denominated. Three architectural families dominate the design space. *Fiat-backed* stablecoins (USDT, USDC) peg through off-chain reserves held by a custodian — operationally simple, but reintroducing the centralized trust that blockchains were built to remove. *Algorithmic* stablecoins, exemplified by TerraUSD prior to its May 2022 collapse, attempt to maintain the peg through programmatic supply contraction backed only by a sister volatility token — a design that has failed catastrophically and repeatedly under stress [1, 9]. *Crypto-collateralized* stablecoins such as MakerDAO's DAI [3] and Liquity's LUSD [4] occupy the middle path: users overcollateralize with on-chain assets, mint a stable token against the position, and are subject to liquidation if collateral value falls below a protocol-defined floor.

The crypto-collateralized design preserves decentralization but introduces a critical dependency: the protocol must read the market price of the collateral asset from an external source to determine whether positions remain solvent. This dependency is the protocol's largest attack surface. On October 11, 2022, Mango Markets — a Solana-based perpetuals exchange — lost approximately \$114 million when an attacker manipulated the on-chain price of the MNGO collateral token across thinly-traded venues, then borrowed against the inflated position before the price could correct [6, 7]. The attack succeeded because the protocol consumed an oracle price within the same transaction window in which the manipulation occurred.

This paper presents *MiniVault*, a minimal CDP-style stablecoin system built to demonstrate the canonical crypto-collateralized design at academic scale, and contributes one defensive feature targeting the Mango-class attack pattern: a *same-block circuit breaker* embedded in the oracle adapter, which rejects price readings whose intra-block movement exceeds a 5% threshold. The system is implemented, tested, deployed to Sepolia with source verified on Etherscan, and exposed through a web frontend. My contributions are (i) a three-contract reference implementation, (ii) the circuit-breaker mechanism, and (iii) a

discussion of how EVM transaction semantics constrain its design.

2. System Design

MiniVault is composed of three Solidity 0.8.24 contracts deployed independently and wired at construction time. Figure 1 sketches the data flow.

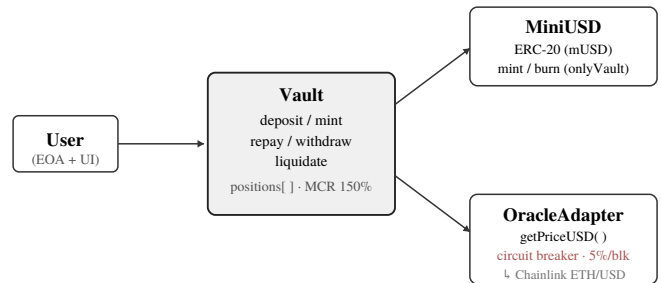


Fig. 1. MiniVault architecture. Vault is the only contract authorized to mint/burn MiniUSD and the only consumer of the OracleAdapter's guarded feed.

MiniUSD is a minimal ERC-20 token (name MiniUSD, symbol mUSD, 18 decimals) inheriting from OpenZeppelin's audited ERC20 base [11]. Mint and burn are gated by an `onlyVault` modifier; the authorized vault address is set exactly once after deployment to resolve the circular constructor dependency between the token and the vault. This one-shot configuration is acceptable for an academic deployment but would be replaced by a more elaborate governance pattern in production.

OracleAdapter wraps Chainlink's Sepolia ETH/USD aggregator (0x694A...5306) [5] and exposes two functions: a state-changing `getPriceUSD()` used by the Vault during mint and withdraw checks, and a view-only `peekPriceUSD()` used by the frontend for display. The state-changing variant performs three guards: a positive-value check on the raw feed answer, a staleness check (reverting if `updatedAt` is older than one hour), and the same-block circuit breaker described in Section 3. The result is

scaled from the feed's native 8 decimals to 18 decimals to match the stablecoin's precision.

Vault stores each user's position as a (collateralETH, debtUSD) pair and exposes the five user-facing actions: `depositCollateral`, `mintStablecoin`, `repay`, `withdrawCollateral`, and `liquidate`. The minimum collateralization ratio is fixed at 150% — a setting comparable to MakerDAO's ETH-A vault [3] but stricter than Liquity's 110% minimum, which relies on a stability pool for liquidations rather than open auctions [4]. Liquidation is implemented in its simplest form: any caller may liquidate an underwater position by burning mUSD equal to the borrower's debt and receiving the entire collateral as compensation. This single-transaction seize model is intentionally cruder than Maker's dutch auctions or Liquity's stability pool, both of which are discussed as future work.

3. Threat Model and the Circuit Breaker

The threat model assumes an adversary with capital sufficient to manipulate the price of the collateral asset on the venues fed into the oracle, and with the ability to execute a complete attack within a single Ethereum transaction. The latter condition is satisfied by any attacker who can use a flash loan — an atomic, single-transaction borrow-and-repay primitive native to DeFi [7]. The Mango Markets incident of October 2022 [6] is the canonical instance: the attacker pumped the price of the MNGO collateral token on thin spot venues, the protocol's oracle consumed the manipulated quote, and the attacker borrowed against the resulting inflated position — all atomically and irreversibly.

The defining property of this attack class is that the manipulation must be visible to the consuming protocol *within the same block* as the exploit. The circuit breaker exploits this property directly: the adapter records the price and block number of every successful read, and on each subsequent read in the same block, it computes the basis-point delta against the previously seen price. If the delta exceeds 500 bps (5%), the function reverts with `CircuitBreakerTripped`, ensuring the Vault never observes the manipulated value. The core check is shown below.

```
if (lastPrice != 0 && block.number == lastBlock-
Seen) {
    uint256 diff = price > lastPrice
        ? price - lastPrice : lastPrice - price;
    uint256 bps = (diff * 10_000) / lastPrice;
    if (bps > MAX_PRICE_MOVE_BPS) {
        tripped = true;
        revert CircuitBreakerTripped();
    }
}
```

An EVM-specific subtlety surfaced during testing and merits explicit discussion. The assignment `tripped = true` is written to storage in the same transaction that subsequently reverts. EVM rollback semantics atomically undo all storage modifications performed by a reverted transaction, so the `tripped` flag never durably transitions to `true` in normal operation. The user-visible security property is nonetheless preserved: the manipulated price is never returned to the Vault, since the call reverts before propagating. However, the breaker does not provide a persistent *cool-down* across blocks — each block is evaluated independently. A production design would split detection from the revert, for example by recording the trip in a separate guard contract whose state-changing call cannot be rolled back by the consumer's revert, or by encoding the trip into a non-reverting branch that the Vault then queries.

The breaker's threshold trades false positives for sensitivity. Legitimate flash-crash conditions — the March 2020 "Black Thursday" event on Ethereum, for instance — can produce single-block moves in the same magnitude as deliberate manipulation. A 5% threshold accepts that the Vault will refuse to mint during such windows, prioritizing solvency over availability. In production, the threshold would itself be a governance parameter.

4. Implementation, Deployment, and Evaluation

The protocol is implemented in Solidity 0.8.24 with the optimizer enabled at 200 runs. Tests use Hardhat 2.28, Mocha, Chai, and ethers v6, with a `MockAggregator` stub providing deterministic prices and a single-transaction `OracleTrigger` helper used to drive the same-block branch of the circuit breaker without relying on automine manipulation. The frontend is a Vite + React + Tailwind single-page application using ethers v6 in the browser. The full test suite — 33 cases spanning metadata, access control, oracle guards, the circuit breaker, the deposit-mint-repay-withdraw round trip, ratio enforcement on both mint and withdraw, and liquidation under price drops — passes in 532 ms.

All three contracts are deployed and verified on Ethereum Sepolia (Table I). A representative `mintStablecoin` transaction was executed at block 10,828,644 on May 10, 2026, minting 1 mUSD from the deployer's address and confirming the live oracle-mediated path end to end. The live demo is available at ezreal222.github.io/minivault.

Contract	Sepolia Address
MiniUSD	0x1EaE3d1aDE6b566B8A2a0Cd6857100D6a9eecdaf
OracleAdapter	0x5C9f2C82E3De483dcc070a01b827001D-Cd3875cb
Vault	0xD642bffff61872c-C899594477A93d7cFEAE74CE6A
Chainlink ETH/USD	0x694AA1769357215DE4FAC081bf1f309aD-C325306

Table I. Deployed contract addresses on Sepolia. All source code verified on Etherscan.

5. Limitations and Future Work

- No stability fee or interest accrual on debt; in production this is essential for protocol revenue and to discourage indefinite borrowing.
- Single collateral asset (ETH only); a multi-collateral design would require per-asset risk parameters and isolation modes.
- Liquidation seizes the borrower's entire collateral. Production systems use partial liquidations (Aave), open dutch auctions (Maker), or stability-pool offsets (Liquity) [3, 4].
- The breaker's tripped flag is non-persistent due to EVM rollback semantics (Section 3); a future iteration would externalize the trip state into a separate, non-reverting guard contract.
- The oracle is a single Chainlink feed; a production deployment would use a median over multiple feeds (Chainlink + Pyth + a TWAP).
- Parameters (MCR, breaker threshold, staleness window) are hardcoded; a governance token and timelock would expose them as adjustable.

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References

1. S. Nakamoto, *Bitcoin: A Peer-to-Peer Electronic Cash System*, 2008.
2. V. Buterin, *A Next-Generation Smart Contract and Decentralized Application Platform (Ethereum Whitepaper)*, 2014.
3. Maker Foundation, *The Maker Protocol Whitepaper*, 2020.
4. R. Lauko and R. Pardoe, *Liquity: Decentralized Borrowing Protocol*, rev. 0.3, Liquity AG, Apr. 2021.
5. S. Ellis, A. Juels, and S. Nazarov, *ChainLink: A Decentralized Oracle Network*, v1.0, Sept. 2017.
6. U.S. Securities and Exchange Commission, *SEC Charges Avraham Eisenberg with Manipulating Mango Markets' Governance Token to Steal \$116 Million of Crypto Assets*, Press Release 2023-13, Jan. 2023. sec.gov/newsroom/press-releases/2023-13
7. P. Daian *et al.*, "Flash Boys 2.0: Frontrunning, Transaction Reordering, and Consensus Instability in Decentralized Exchanges," in *IEEE Symposium on Security and Privacy (S&P)*, 2020.
8. S. M. Werner *et al.*, "SoK: Decentralized Finance (DeFi)," in *Proc. 4th ACM Conference on Advances in Financial Technologies (AFT)*, 2022, pp. 30–46.
9. A. Klages-Mundt, D. Harz, L. Gudgeon, J. Liu, and A. Minca, "Stablecoins 2.0: Economic Foundations and Risk-Based Models," in *Proc. AFT*, 2020, pp. 59–79.
10. L. Gudgeon, S. M. Werner, D. Perez, and W. J. Knottenbelt, "DeFi Protocols for Loanable Funds: Interest Rates, Liquidity and Market Efficiency," in *Proc. AFT*, 2020.
11. OpenZeppelin, *OpenZeppelin Contracts v5.0 Documentation*, 2024. docs.openzeppelin.com/contracts/5.x
12. Chainlink, *Price Feed Contract Addresses — Sepolia Testnet*, 2026. docs.chain.link/data-feeds/price-feeds/addresses